

are created for augmentation, virtual analysis or simple replication using a 3D printer. These virtual simulations, either manually designed through CAD or acquired using scanners, are the necessary first step towards beginning any 3D printing project.

Step 2: The next step on the 3D printing journey is the conversion of the CAD-based models and designs into a language format that's compatible with that of 3D printers – the STL format. The STL format, or 'standard tessellation language' format, is the current industry standard that was developed for the use of 3D printers. It was originally created in 1987 for use on stereolithography apparatus machines (We'll come back to this and other types of 3D printers at the end of this chapter). Although STL format files are the primary standard for 3D printers, a small group of other assorted proprietary



As with any machine a little hands-on adjustment is always necessary

file formats such as ZPR and ObjDF which require specialised software can also do the job. However, the significant majority of professional as well as Open Source software support STL, so new users don't need to worry about multiple conversions.

Step 3: The next step will determine how the 3D printer will interpret the

STL file design. This is where "Print Properties" comes into the picture. In the same way that we adjust printer properties such as horizontal or vertical orientation when we print a 2D document, we can adjust properties such as size and print orientation of an STL file when printing a 3D design.

Step 4: This step varies according to the type of the printer. Once the STL file is ready for printing, the machines need to be checked for the required materials and placement configurations, just as a paper printer needs to be checked for ink and tray alignment. In the case of 3D printing, the types of machines vary greatly based on their printing techniques, and accordingly require different types of materials to work with. This includes polymers, binders, adhesives and powders. In addition to this, the placement of the base tray or chemical solution base also needs to be adjusted.